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Integrated locking mechanism for drive blanks

Abstract

A drive blank comprises a front member having a first end, a second end, a back face, and an aperture formed between the first end and the second end. The drive blank further has a first side member and a second side member each extending perpendicularly from the first end and the second end of the front member, respectively. The drive blank has a rear member extending parallel to the front member and between the first and second side members. A locking element is formed integral to the front member. The locking element comprises a looping portion having a first end and a second end and a push button formed at the second end of the looping portion and disposed in the aperture. The push button has a locking catch, releasably coupled to a latch element of a handle, to lock the handle in a closed position.

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Background/Summary

BACKGROUND

(1) Computer systems may include one or more media drives such as, solid state drives (“SSDs”), hard disk drives (“HDDs”), CD ROM drives, or DVD drives. One or more receiving locations (e.g., drive bays) can be included in the drive cage of a computer for receiving these media drives. The media drives can either be directly mounted within those receiving locations, or may be first received in a carrier, which is itself receivable in the receiving location. A drive carrier may be used, for example, in a computing system (e.g., a server, a high-performance-compute system, a data storage appliance, a converged or hyperconverged system, or other computing systems) to facilitate installation and uninstallation (e.g., hot-swapping) of a media drive to the computing system.

(2) A drive cage of a computing system may include multiple drive bays each of which may be filled with a drive carrier (which may be carrying a media drive) or may remain vacant (i.e., not filled with a drive carrier). Further, the vacant bays in the drive cage of a computing system may be filled with drive blanks. Drive blanks are devices that are configured to occupy a drive bay in a similar manner as a drive carrier, but which are not configured to carry a media drive. Drive blanks are generally used to fill bays that do not have drive carriers installed therein, as opposed to leaving those bays vacant. One reason to use drive blanks is to ensure proper airflow through the drive cage. Generally, air will be pulled into or exhausted from the drive cage at least in part via the drive bays, with the air flowing into or out of gaps between the devices installed in these bays. When all

of the bays are occupied, the impedance to the airflow through each bay is essentially the same because the gaps in the bays are all essentially the same, and thus the air flowing through the bays will be distributed uniformly among the bays. This helps to ensure more uniform cooling of the devices in the bays, and also the other components of the computing system by directing flow of cooling air through all drive bays of the computing system. However, if one or more bays are left vacant, then the impedance through those vacant bays will be much lower than other bays, and thus more air will flow through the vacant bays, which lowers the airflow through the occupied bays, degrading the cooling of the devices in those bays, and also causes non-uniform airflow through the bays as a whole, which may degrade the cooling of the other components in the system. Blanks can be used to prevent this, as the blanks can be inserted in the vacant bays and cause those bays to have a similar impedance as the bays filled with drive carriers, thus preventing excessive flow of cooling air through the vacant drive bays. Thus, the drive blanks aid in directing the flow of cooling air through the drive cage of the computing system by allowing even distribution of cooling air between vacant bays and bays filled with a drive (or a drive carrier). Another reason to use drive blanks is to help mitigate electromagnetic interference (EMI) due to certain types of EM radiation emitted by the devices inside the drive cage. Openings in the drive cage, such as the drive bays, act like windows through which EMI causing radiation can escape, and thus the drive carriers include EMI mitigation features, such as spring fingers that electrically connect with the drive cage, to reduce the emission of EMI causing radiation through the bay. However, if a bay is left vacant, it can be a source of EMI leakage. Thus, drive blanks often include EMI mitigation features that may be similar to those of drive carriers, so that the drive blanks can be inserted into a vacant bay to reduce the EMI leakage causing radiation through that bay. Drive blanks also prevent any unintentional mishandling of the vacant drive bays. In particular, when a drive bay is vacant, a user of the computing system may unintentionally insert another electronic component, other than a drive, in the vacant drive bay. This may cause the computing system to malfunction. If a vacant drive bay is filled with a blank, chances of such unintentional mishandling may be avoided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is best understood from the following detailed description when read with the accompanying figures, wherein:
- (2) FIG. 1 is a front isometric view illustrating an drive blank, according to an example of the present subject matter;
- (3) FIG. 2 is a rear isometric partial view of the drive blank of FIG. 1;
- (4) FIG. 3 is a top isometric sectional view illustrating a portion of the drive blank of FIG. 1, with section taken along 3-3 in FIG. 2;
- (5) FIG. 4A is a cross section view illustrating a portion of the drive blank of FIG. 1, with the section taken along 4-4 in FIG. 1;
- (6) FIG. 4B is a cross section view illustrating a portion of the drive blank of FIG. 1, with the section taken along 4-4 in FIG. 1;
- (7) FIG. 5 is a top isometric view of a handle according to an example of the present subject matter;
- (8) FIG. 6 is a front isometric view of the drive blank of FIG. 1;
- (9) FIG. 7A is an isometric view of a drive cage of a computing system fitted with the drive blank of FIG. 1;
- (10) FIG. 7B is a partial isometric view of a portion of the drive cage of the computing system of FIG. 7A; and
- (11) FIG. 8 is a method of installing a drive blank in a drive cage of a computing system, according to an example of the present subject matter.

(12) It is emphasized that in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion or illustration.

DETAILED DESCRIPTION

(13) Illustrative examples of the subject matter claimed below are disclosed. In the interest of clarity, not all features of an actual implementation are described in this specification. It will be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions may be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort, even if complex and time-consuming, would be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

(14) As used herein, the article “a” is intended to have its ordinary meaning in the patent arts, namely “one or more.” Herein, the term “about” when applied to a value generally means within the tolerance range of the equipment used to produce the value, or in some examples, means plus or minus 10%, or plus or minus 5%, or plus or minus 1%, unless otherwise expressly specified. Further, herein the term “substantially” as used herein means a majority, or almost all, or all, or an amount with a range of about 51% to about 100%, for example. Moreover, examples herein are intended to be illustrative only and are presented for discussion purposes and not by way of limitation.

(15) As used herein, to “provide” an item means to have possession of and/or control over the item. This may include, for example, forming (or assembling) some or all of the item from its constituent materials and/or, obtaining possession of and/or control over an already-formed item.

(16) Drive carriers are used to configure media drives, such as SSDs, as FRUs, allowing the drives to be hot swapped into a computing system. Drive carriers are therefore defined by the physical form of the SDD, or drive, which they are configured to carry and are standardized by an industry form factor (e.g., an SDD form factor). An Enterprise & Data Center Standard Form Factor (EDSFF), for example, is designed for use in data centers in servers with storage devices such as SSDs. As a form factor, it defines specifications such as the dimensions and electrical interfaces storage devices should have, to ensure data center operators, server manufacturers, and SSD manufacturers, can make products that work with products from multiple manufacturers. As used herein, an EDSFF drive therefore refers to a drive that has a form factor specified in one of the standards in the EDSFF family of standards, including, but not limited to, a form factor specified in SFF-TA-1006, SFF-TA-1007, or SFF-TA-1008, and an EDSFF drive carrier is a drive carrier configured to carry and support an EDSFF drive.

(17) Due to the small size of emerging EDSFF technologies, drive carriers designed to support EDSFF drives (EDSFF drive carriers) are significantly smaller than those designed for previous drive technologies (e.g., HDDs, SSDs conforming to M.2 or U.2 specifications, etc.) Similarly, drive blanks designed to fill vacant EDSFF drive bays (EDSFF drive blanks) are also significantly smaller than drive blanks developed for previous drive technologies. The small size of such EDSFF drive blanks raises various new issues. Given the larger size of drive blanks for previous drive technologies, in the past, it was possible to use a squeeze latch to retain and release such drive blanks. As the size of the drive carrier (and corresponding drive blank) decreases, however, it may not be feasible to use a squeeze latch to retain and release an EDSFF drive blank from the drive bay due to space constraints of installing such squeeze latch.

(18) Moreover, in the past, it was possible to use handle based latching mechanisms to lock a drive blank in place inside a drive cage of a computing system. Such handle based locking mechanisms used in the past, generally had a handle which would be locked by a locking mechanism separately mounted on the drive blank. However, for EDSFF drive blanks, there may be insufficient space available for separately mounting the locking mechanism of the handle on the drive blank. Thus, it

may not be feasible to lock the EDSFF drive blank using such systems that use a locking mechanism separately mounted on the drive blank. Also, attaching and tooling such locking mechanism on the EDSFF drive blank may be complex due to space constraints of the EDSFF drive blank. Further, such handle-based locking mechanisms for previous drive technologies generally employ multiple disjoint components to be installed (e.g., mounted) on the drive blank or drive bay for securing the drive blank inside the drive bay. For EDSFF drive blanks, however, installing such multiple disjoint components may be complex and time consuming due to the compactness of the EDSFF drive blanks. Further, installing such multiple disjoint components may also involve additional tooling costs.

(19) To address the challenges associated with drive blanks for small drives, including EDSFF drives, the present disclosure contemplates an improved drive blank and assembly that, among other things, provide a compact locking system that is integral to the drive blank, such that the locking system is formed as a single piece along with the drive blank. Also, since the locking system is not separate from the drive blank, it need not be installed or mounted on the drive blank, thereby reducing assembly and tooling costs. In some implementations, the drive blank as disclosed herein comprises a locking element that is integral to a front member of the drive blank frame. The locking element is responsible for locking a handle that, in a closed position, holds the drive blank inside a slot in a drive cage of a computing system (e.g., a rack). The compact design and placement of the locking element allows the locking element to be formed to be integral to the drive blank frame.

(20) FIG. 1 is a front isometric view illustrating a drive blank **100** according to an example of the present subject matter. The drive blank **100** comprises a front member **102** having a first end **102-a** and a second end **102-b**. The front member **102** is a frame element of the drive blank **100** that is designed to face a user of the drive blank **100** when the drive blank **100** is inserted in a bay of a drive cage of a computing system, such as a bay **707** of drive cage **700** shown in FIG. 7A. The drive blank **100** further comprises a first side member **104** and a second side member **106**. The first side member **104** extends perpendicularly from the first end **102-1** of the front member **102** and the second side member **106** extends perpendicularly from the second end **102-b** of the front member **102**.

(21) The drive blank **100** further comprises a rear member **108** extending parallel to the front member **102** and between the first side member **104** and the second side member **106**. Thus, in this example, the drive blank **100** has a substantially rectangular frame structure. The front member **102**, the first side member **104**, the second side member **106**, and the rear member **108**, are connected such that together they form a frame structure that can fill a vacant drive bay in a drive cage of a computing system. In some examples, the drive blank **100** is designed to fit in a bay of a drive cage of a computing system for housing a media drive for an electronic device (e.g., a computing device), including, but not limited to an SSD. It should be understood, however, that in other examples the drive blank may have different dimensions, shapes, and that various structural details may differ from those shown in the example of FIG. 1. Moreover, in some implementations the drive blank **100** may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated.

(22) In some examples, the drive blank **100** may be inserted in a bay of a drive cage of a computing system to prevent improper cooling and thermal damage. During operation, bays of the computing system are populated either with an SSD component or a drive blank, such as the drive blank **100** of FIG. 1. The drive blank **100** comprises a locking element **110** integral to the front member **102**. The locking element **110** is shown in further details in FIG. 3.

(23) In various implementations, the front member **102**, the first side member **104**, and the second side member **106** are formed from a plastic material. The front member **102**, the first side member **104**, and the second side member **106** may for example, be integrally formed together from a polycarbonate. Furthermore, in some implementations, the front member **102**, the first side member

104, and the second side member **106** are formed from polycarbonate/acrylonitrile butadiene styrene (PC-ABS).

(24) FIG. 2 is a rear isometric partial view of the drive blank **100** of FIG. 1. In the view of FIG. 2, the drive blank **100** is shown as it appears when viewed in a direction from the rear member **108** (not shown in FIG. 2) towards the front member **202**. With reference to FIG. 2, the front member **102** has the first end **102-a**, the second end **102-b**, a front face **202**, and a back face **204** opposing the front face **202**. The back face **204** refers to a surface of the front member **102** that faces the rear member of the drive blank **100**. Further, the front member **102** has an aperture **206** between the first end **102-a** and the second end **102-b**. The aperture **206** refers to a through hole formed on the front member **102**. At least a portion of the locking element **110** is disposed in the aperture **206**. The locking element **110** is integral to the front member **102**. The locking element **110** comprises a looping portion **208** as shown in FIG. 2. The portions of the locking element **110** are shown in further details in FIG. 3.

(25) FIG. 3 is a top perspective sectional view illustrating a portion of the drive blank **100** of FIG. 1. The drive blank **100** comprises the locking element **110** integral to the front member **102**. The locking element **110** comprises the looping portion **208** and a push button **302**. The looping portion **208** has a first end **304** and a second end **306** and the push button **302** is formed at the second end **306** of the looping portion **208**. The front member **102** has a first wall section **308** and a second wall section **310**. The first wall section **308** and the second wall section **310** are on either sides of the aperture **206** (shown in FIG. 2) and a majority of the locking element **110**, including the push button **302**, is located between the two wall sections, i.e., the first wall section **308** and the second wall section **310**. The first end **304** of the looping portion **208** is integrally coupled to the back face **204** of the front member **102** (i.e., a back face of the first wall section **308**). The looping portion **208** refers to an elastically bendable curved portion of the locking element **110** that protrudes generally in a direction that is perpendicular to the extending direction of the front member **102** and towards a direction of a rear member, illustrated by arrow **312**, of the drive blank **100**. In some examples described herein, the looping portion **208** is a U-shaped loop extending between the back face **204** of the front member **102** and a back surface of the push button **302**. That is, the loop portion **208** may begin by extending from the first wall section **308** generally in the direction **312**, then begin to change its direction of extend in curved manner (curving clockwise in the view of FIG. 3), until the loop has curved through approximately 180 degrees and is ultimately extending generally in a direction that is opposite the direction **312** when the loop portion **208** joins with the push button **302**. Further, the push button **302** has a tactile surface **314** protruding from the front face **202**, opposite to the back face **204**, of the front member **102**. The front face **202** of the front member **102** is a surface of the front member **102** facing the user when the drive blank **300** is installed in a drive cage of a computing system. Further, as shown in FIG. 3, the looping portion **208** comprises a flared ridge **316** extending along a perimeter of the looping portion **208** between the first end **304** and the second end **306** of the looping portion **208**. The flared ridge **316** is an extension on the surface profile of the looping portion **208** along its perimeter that may add to the tensile strength of the looping portion **208**. Note that in FIG. 3 the flared ridge **316** is mostly visible near the second end **306** of the looping portion **208**, but as shown in FIG. 2 the flared ridge **316** also continues around the looping portion **208** to the first end **304** thereof. The flared ridge **316** tends to widen or flare out more near the first and second ends **304** and **306**, which corresponds to relatively straight sections of the looping portion **208** where the looping portion **208** is extending generally along the direction **312** or opposite to the direction **312**, and tends to be much narrower along the curved sections of the looping portion **208**, conforming to the curved shape of the looping portion **208** in these regions. The wider portions of the flared ridge **316** along the straighter sections lends these sections increased rigidity and strength. At the first end **304**, this increased rigidity and strength can help to reduce bending near the first end **304**, which can help to reduce stress on the junction between the looping portion **208** and the wall section **308**. At the second end **306**, the

increased rigidity and strength can help prevent bending of the push button **302** relative to the looping portion **208**. In other words, in addition to strengthening the looping portion **208**, the flared ridge **316** may help to confine the bending to predominantly portions that are intended to bend (e.g., the curved section) while reducing bending in portions that is desired to not bend.

(26) FIGS. **4A** and **4B** are cross section views illustrating a portion of the drive blank **100** of FIG. **1** in two different positions of a handle **402**. In FIG. **4A**, the handle **402** is in an open position and in FIG. **4B** the handle is in a closed position. The handle **402** is pivotably coupled to the first end **102a** of the front member **102** (and/or to the side member **104**) by a pivot pin **404**, which allows the handle **402** to move between the open and closed positions. The handle **402** is used to lock the drive blank **100** into the drive cage of a computing device (in the closed position of the handle) or to release the drive plank **100** therefrom (in the open position of the handle). This is accomplished via interaction between a locking pawl **508** and the drive cage, which will be described in greater detail below with respect to FIG. **5**. The pin **404** may have a spring **406** attached thereto. The structure and arrangement of the pin **404** and the spring **406** are elaborated further in conjunction with the description of FIG. **6**. The handle **402** so fastened to the drive blank **100** may be rotatable pivotally around the pin **404**. The spring **406** may engage with the front member **102** and the handle **402** so as to exert a spring force thereon that biases the handle **402** toward the open position. That is, when the handle **402** is moved from the open position of FIG. **4A** to the closed position as shown in FIG. **4B**, the spring **406** gets compressed, generating a restorative spring force urging the handle **402** towards the open position.

(27) As shown in FIGS. **4A** and **4B**, the handle **402** has a latch element **408** at an end opposite to the end of the handle **402** that is pivotally secured to the drive blank **100** via the pin **404**. The latch element **408** is a chamfered extension from an edge of the handle **402**, and comprises an angled (sloped) lead-in surface **408a** on a rearward side thereof and an interlocking surface **408b** on a frontward side thereof. In the open position of the handle **402**, as shown in FIG. **4A**, the latch element **408** of the handle **402** does not interact with any other elements of the drive blank **100**. Once the drive blank **100** is inserted in a drive cage of a computing system, the handle **402** may be rotated, from the open position (as shown in FIG. **4A**), in a direction of the arrow **410**, towards the front member **102** of the drive blank **100**. Movement of the handle **402** from the open position towards the front member **102** may result in the spring **406** getting compressed. When the handle **402** is sufficiently close to the front member **102**, the angled lead-in surface **408a** of the latch element **408** on the handle **402** may engage with a lead-in surface **412a** of a locking catch **412** formed on an end of the push button **302**, as shown in FIG. **4A**. The locking catch **412** is formed at the end of the push button **302** that is away from the second end **102-b** of the front member **102**. As the handle **402** is further rotated closer to the front member **102** in the direction of the arrow **410**, the angled lead-in surface **408a** of the latch element **408** may slide against the lead-in surface **412a** of the locking catch **412**, imparting a force urging the locking catch **412** in the direction **416** (towards second end **102-b**) which causes the looping portion **208** of the locking element **110** to bend to accommodate the motion of the locking catch **412**. Once the angled lead-in surface **408a** of the latch element **408** has slid past the lead-in surface **412a** of the locking catch **412**, the force exerted by the latch element **408** on the locking catch **412** ceases and thus there is no longer any force holding the looping portion **415** in the bent state. Accordingly, the elastic restoring force generated by the bending of the looping portion **415** now causes the looping portion **415** to elastically rebound to its original position, along a direction of the arrow **418**, which causes the latch element **408** to move back towards its original position. The latch element **408** moving back to its original position allows the interlocking surface **408b** of the latch element **408** to interlock against the interlocking surface **414** of the locking catch **412**. That is, the locking catch **412** moves back to its original position thereby locking the latch element **408** behind it and consequently retaining the handle **402** in the closed position, as shown in FIG. **4B**. The closed position refers to a position of the handle **402** (as shown in FIG. **4B**) in which the latch element **408** of the handle **402**

interlocks with the locking catch **412**, such that the handle **402** retains the drive blank **100** inside a drive cage of a compute system. In the closed position of the handle **402**, the spring **406** may be in a fully compressed state and may exert a force on the handle **402** in a direction of the arrow **420**. Since, the locking catch **412** interlocks with the latch element **408** in the closed position, the handle **402** is prevented from moving away towards the open position due to the force exerted by the spring **406**. As a result, the drive blank **100** is also held inside the drive cage by the handle **402** in the closed position.

(28) To release the handle **402** from the closed position to the open position, a force may be applied to the push button **302** which causes the looping portion **208** to elastically bend, moving the push button **302** in a direction towards the second end **102-b** of the front member **102** away from the latch element **408**. As the push button **302** moves away from the latch element **408**, the interlocking between the latch element **408** and the locking catch **412** may no longer be there. Thus, the force exerted by the compressed spring **406** in the closed position of the handle **402** may rotate the handle **402** in the direction of arrow **422** as shown in FIG. **4B**, away from the front member **102** of the drive blank **100** thereby moving the handle **402** back to the open position. In some examples, the push button **302** may be subjected to an amount of force ranging from about 0.5 pounds to about 0.6 pounds to deflect the push button for about 0.65 millimeter to about 0.70 millimeter such that the handle **402** is moved from the closed position to the open position along the direction of the arrow **422**. Since, the locking element **110** does not employ a separate button and button spring, additional parts, labor to assemble such additional parts, and a higher development cost arising from tooling of the additional parts may be avoided. This may provide a locking mechanism for the drive blank that is cost effective in addition to being compact as per the EDSFF form factor requirements.

(29) Further, as shown in FIG. **4B**, the looping portion **208** may have a thickness “a” at a portion where the curvature/bend of the looping portion **208** is the greatest. In some examples, the thickness “a” may be in a range between about 2 millimeters (mm) to about 3 mm. The push button **302** may have a length “b”. In some examples, the length “b” may be in a range between about 4 mm to about 6 mm. The handle may have a length “c”. In some examples, the length “c” may be in a range between about 40 mm to about 50 mm.

(30) FIG. **5** is an example top isometric view of the handle **402** of FIG. **4**. The handle **402** comprises a first end **502** to be pivotally secured to an end of a first side member of a drive blank, such as the first side member **104** of the drive blank **100**. The first end **502** has a through hole **504** through which a pin may pass and act as a pivot. The handle **402** when assembled with the drive blank may rotate about the first end **502**. The handle **402** has a second end **506**, opposite to the first end **502**, bearing the latch element **408**. The latch element **408** is a component in the handle **402** that engages with a locking catch, such as a locking catch **412** of FIG. **4**, in the locking element **110** to lock the handle **402** in the closed position. The handle **402** comprises a locking pawl **508** at the first end **502**. The locking pawl **508** is a protrusion to movably engage with a side rail of a drive cage of a computing system in which the drive blank is to be inserted. A locking pawl engages with a fixed component to prevent movement of another component in one or more directions. In some examples described herein, the locking pawl **508** formed on the first end **502** of the handle **402** may engage with a siderail of a drive cage, when the handle **402** is in the closed position, to prevent removal of the blank **100** from the drive cage. Specifically, the pawl **508** and a front portion of the handle **402** may define a slot **508c** therebetween and the side rail of the drive cage may be received within this slot **508c** when the blank **100** is installed in the drive cage and the handle **402** is in the closed position; furthermore, in this state a forward facing surface **508a** of the pawl **508** blocks the side rail to prevent translational motion of the drive blank **100** out of the drive cage. In other words, when the handle **402** is in the closed position the locking pawl **508** prevents removal of the drive blank **100** from the drive cage. In some examples, the pawl **508** may also assist in fully installing the drive blank **100** in the drive cage; for example, the drive blank **100** may be inserted part way

into the bay and then the handle is rotated from open position to closed position causing the pawl **508** to engage the drive cage, and as the pawl **508** is rotated it acts as a cam to force the drive blank **100** to move the rest of the way into the fully installed position. In addition, a rearward surface **508b** of the pawl **508** may engage the front member **102** as the handle **402** is rotated toward the open position to limit rotational movement of the handle **500 402** beyond an the open position. (31) FIG. **6** shows an isometric view of a drive blank **100** according to an example of the present subject matter. In FIG. **6**, a partial view of the first side member **104** and a partial view of the second side member **106** is shown.

(32) The first side member **104** comprises a pair of retention holes **602** for the handle **402**, one disposed on a top side of the handle **402** and another disposed on a bottom side of the handle **402**. The handle **402** has a first end **604** and a second end **606**. The first end **604** of the handle **402** is pivotally secured to an end of the first side member **104** via the retention holes **602**. The handle **402** has a bore **504** that is aligned with the retention holes **602**, and the pin **404** passes through the bore **504** and the retention holes **602** and thereby fastens the handle **402** to the first side member of the drive blank **100** in a manner that allows pivoting of the handle **402**. The handle **402** has the second end **606**, opposite to the first end **604**, bearing the latch element **408**.

(33) Further, the retention holes are **602** positioned at the end of the first side member **104** that is adjacent to the first end **102-a** of the front member **102**. The drive blank **100** further comprises the pin **404** to pivotally secure the handle **402** in the retention holes **602**. The spring **406** is configured to bias the handle **402** in an open position.

(34) Further, as shown in FIG. **6**, the front member **102** comprises a first edge **608** between the first end **102-a** and the second end **102-b** of the front member **102**. The front member **102** also has a second edge **610**, opposite to the first edge **608**. The first edge **608** has a first set of outward extensions **610** and the second edge **610** has a second set of outward extensions **612**. The first and second set of outward extension **620** and **622** together form a channel therebetween to hold the handle **402** in the closed position. That is in the closed position of the handle **402**, the handle **402** may be fitted between the first and second set of outward extension **610** and **612**.

(35) FIG. **7A** shows a drive cage **700** of a compute system fitted with the drive blank **100** of FIG. **1**. The drive cage **700** comprises bays **707**, as described above (only one is labeled in FIG. **7A**, but multiple are shown). The drive blank **100** is fitted with the handle **402** with the first end **604** and the second end **606**. The handle **402** may movably engage with a siderail **704** of the drive cage **700**. The connection between the handle **402** and the drive cage **700** is shown in greater detail in FIG. **7B**. FIG. **7B** shows a magnified view of the circular portion **702** identified in FIG. **7A**. As shown in FIG. **7B**, the handle **402** comprises the locking pawl **508** at the first end **604**. The locking pawl **508** extends into a cut-out **705** or recess in the drive cage **700** to movably engage with a side rail **704** of the drive cage **700** in which the drive blank **100** is inserted. When the drive blank **100** is installed in the drive cage **700** and the handle **402** is closed, the side rail **704** is received in the slot **508c** defined between a front of the handle **402** and the pawl **508**, with the side rail **704** contacting or blocking the forward facing surface **508a** of the pawl **508** to lock the drive blank **100** in the drive cage **700**.

(36) FIG. **8** shows a method **800** of installing a drive blank in a computing system, according to an example of the present subject matter. The drive blank may be similar to the drive blank **100** as illustrated in any of the FIGS. **1-6** above. The method **800** comprises at block **802**, providing the drive blank. The drive blank comprises a front member having a first end, a second end, a back face, and an aperture formed between the first end and the second end. The drive blank further comprises a first side member extending perpendicularly from the first end of the front member, a second side member extending perpendicularly from the second end of the front member, and a rear member extending parallel to the front member and between the first side member and the second side member. The drive blank comprises a locking element integral to the front member, where the locking element comprises a looping portion having a first end and a second end, the first

end of the looping portion abutting the back face of the front member. In some examples, the looping portion comprises a first end abutting a back face of the front member, a second end, opposite to the first end, and a flared ridge extending, between the first end and the second end, along a perimeter of the looping portion. Further, the drive blank has a push button formed at the second end of the looping portion and disposed in the aperture, the push button having a locking catch, releasably coupled to a latch element of a handle, to lock the handle in a closed position, where the push button is movable towards the rear member to elastically bend the looping portion to release the latch element from the locking catch, such that the handle moves from the closed position to an open position.

(37) The method **800** comprises, at block **804**, inserting the drive blank into a bay of the computing system, such as a bay **707** of the drive cage **700** of a computing system of FIG. **7A**, in a manner that the push button is exposed at a front side of the compute system. The push button is exposed at the front side, such that the push button may be accessed/pushed by a user/operator to release the handle from the closed position to the open position and pull out the drive blank from the drive cage. Once the drive blank is inserted in the drive cage, the method **800** comprises, at block **806**, moving the handle from the open position to the closed position, where the locking catch interlocks with the latch element in the closed position of the handle.

(38) In some examples, the method **800** comprises providing a retention slot for the handle at an end of the front member. Providing the retention slot may refer to forming a slot at an end of the front member. Further, a pin may be inserted in the retention slot to pivotally secure the handle in the retention slot.

(39) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the disclosure. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the systems and methods described herein. The foregoing descriptions of specific examples are presented for purposes of illustration and description. They are not intended to be exhaustive of or to limit this disclosure to the precise forms described. Many modifications and variations are possible in view of the above teachings. The examples are shown and described in order to best explain the principles of this disclosure and practical applications, to thereby enable others skilled in the art to best utilize this disclosure and various examples with various modifications as are suited to the particular use contemplated. It is intended that the scope of this disclosure be defined by the claims and their equivalents below.

Claims

1. A drive blank, comprising: a front member having a first end, a second end, a back face, and an aperture formed between the first end and the second end; a first side member extending perpendicularly from the first end of the front member; a second side member extending perpendicularly from the second end of the front member; a rear member extending parallel to the front member and between the first side member and the second side member; a handle pivotally coupled to the front member or first side member and comprising a latch element; and a locking element integral to the front member, wherein the locking element comprises: a looping portion having a first end and a second end, the first end of the looping portion integrally connected to the back face of the front member; and a push button formed at the second end of the looping portion and disposed in the aperture, the push button having a locking catch, releasably coupled to the latch element of the handle, to lock the handle in a closed position, wherein the push button is movable towards the rear member to elastically bend the looping portion to release the latch element from the locking catch, such that the handle is allowed to move from the closed position to an open position.

2. The drive blank of claim 1, wherein the looping portion comprises a flared ridge extending along a perimeter of the looping portion between the first end and the second end of the looping portion.

3. The drive blank of claim 1, wherein the locking catch interlocks with the latch element in the closed position of the handle.
4. The drive blank of claim 1, wherein the first side member comprises: a pair of retention holes for the handle at an end of the first side member that is adjacent to the first end of the front member; and a spring-loaded pin to pivotally secure the handle in the retention holes.
5. The drive blank of claim 4, wherein the handle comprises: a first end pivotally secured to the end of the first side member; and a second end, opposite to the first end, bearing the latch element.
6. The drive blank of claim 5, wherein the handle comprises a locking pawl at the first end, the locking pawl having a protrusion to movably engage with a side rail of a computing system in which the drive blank is to be inserted.
7. The drive blank of claim 1, wherein the latch element comprise an angled lead-in surface configured to slide against a lead-in surface of the locking catch and to force the locking catch to move aside when the handle is rotated from the open position towards the closed position.
8. The drive blank of claim 1, wherein the front member comprises: a first edge between the first end and the second end of the front member, the first edge having a first set of outward extensions; a second edge, opposite to the first edge, having a second set of outward extensions; and a channel formed between the first set of outward extensions and the second set of outward extensions to receive at least part of the handle in the closed position.
9. The drive blank of claim 1, wherein the push button has a tactile surface protruding from a front face, opposite to the back face, of the front member.
10. The drive blank of claim 1, wherein the drive blank is configured for insertion in a bay of a computing system.
11. The drive blank of claim 10, wherein the push button is accessible from a front side of the computing system with the drive blank being inserted in the computing system.
12. The drive blank of claim 1, wherein the handle has a length of about 40 millimeters (mm) to 50 mm.
13. The drive blank of claim 1, wherein the push button has a length of about 4 mm to 6 mm.
14. The drive blank of claim 1, wherein the push button is configured to deflect about 0.65 mm to about 0.70 mm and release the handle in response to being subjected to a force having a magnitude of about 0.5 pounds to about 0.6 pounds.
15. The drive blank of claim 1, wherein a central axis of the looping portion follows a looping path from the first end of the looping portion to the second end of the looping portion, starting at the back face of the front member and extending from the back face in a first direction towards the rear member and then looping around to extend in a second direction opposite the first direction.
16. An electronic device comprising: a drive cage comprising a plurality of bays each configured to receive a drive carrier; a drive blank configured to be installed in one of the bays in lieu of the drive carrier, the drive blank comprising: a front member having a first end, a second end, a back face, and an aperture formed between the first end and the second end; a first side member extending perpendicularly from the first end of the front member; a second side member extending perpendicularly from the second end of the front member; a rear member extending parallel to the front member and between the first side member and the second side member; a handle to retain the drive blank in the bay, the handle pivotally coupled to the front member or first side member and comprising a latch element; and a locking element integral to the front member, wherein the locking element comprises: a looping portion having a first end and a second end, the first end of the looping portion abutting the back face of the front member; and a push button formed at the second end of the looping portion and disposed in the aperture, the push button having a locking catch, releasably coupled to the latch element of the handle, to lock the handle in a closed position, wherein in the closed position the handle engages the drive cage to lock the drive blank in the drive cage, wherein the push button is movable towards the rear member to elastically bend the looping portion to release the latch element from the locking catch, such that the handle is allowed to move

from the closed position to an open position, and wherein in the open position the handle allows the drive blank to be removed from the drive cage.

17. The electronic device of claim 16, wherein the looping portion comprises a flared ridge extending, between the first end and the second end, along a perimeter of the looping portion.

18. The electronic device of claim 16, wherein the locking catch interlocks with the latch element in the closed position of the handle.

19. The electronic device of claim 16, wherein the push button is accessible from a front side of the electronic device with the drive blank being inserted in one of the bays.

20. A method of installing a drive (SSD) blank in a computing system, comprising: providing the drive blank, the drive blank comprising: a front member having a first end, a second end, a back face, and an aperture formed between the first end and the second end; a first side member extending perpendicularly from the first end of the front member; a second side member extending perpendicularly from the second end of the front member; a rear member extending parallel to the front member and between the first side member and the second side member; and a locking element integral to the front member, wherein the locking element comprises: a looping portion having a first end and a second end, the first end of the looping portion abutting the back face of the front member; and a push button formed at the second end of the looping portion and disposed in the aperture, the push button having a locking catch, releasably coupled to a latch element of a handle, to lock the handle in a closed position, wherein the push button is movable towards the rear member to elastically bend the looping portion to release the latch element from the locking catch, such that the handle is allowed to move from the closed position to an open position; inserting the drive blank into a bay of the computing system such that the push button is exposed at a front side of the computing system; and moving the handle from the open position to the closed position, wherein the locking catch interlocks with the latch element in the closed position of the handle.

21. The method of claim 20, further comprising: providing a retention slot for the handle at an end of the front member; and inserting a spring-loaded pin in the retention slot to pivotally secure the handle in the retention slot, wherein the spring-loaded pin is to bias the handle in the open position.
